



Chapter 4 (Part 1)

Entity Relationship (ER) Modeling

Entity Relationship Model (ERM)

- Basis of an entity relationship diagram (ERD)
- ERD depicts the:
 - Conceptual database as viewed by end user
 - Database's main components (Entities, Attributes, Relationships)

Entity

- An object of interest to the end user (“Person, place or thing about which we want to store info”)
- Entity refers to the **entity set** and NOT to a single entity occurrence

Attributes

- Characteristics of entities
- **Required attribute:** Must have a value, cannot be empty
- **Optional attribute:** Doesn't require value, can be empty
- **Domain** - Set of possible values for a given attribute
- **Identifiers (Primary Keys):** One or more attributes that uniquely identify each entity instance or occurrence.

Underline the PK in the ERD and in a **relational schema**

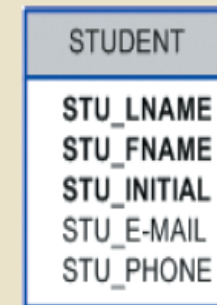
TableName(**KeyAttribute**, Attribute1, Attribute2,.....AttributeK)

Figure 4.1 - The Attributes of the Student Entity: Crow's Foot

FIGURE 4.1 THE ATTRIBUTES OF THE STUDENT ENTITY: CHEN AND CROW' FOOT

Crow's Foot Model

The first 3 attributes are in **BOLD** to indicate that they form the PK



Attributes (cont)

- **Composite identifier:** Primary key composed of more than one attribute
- **Composite attribute:** Attribute that can be subdivided to yield additional attributes (eg. TelNr)
- **Simple attribute:** Attribute that cannot be subdivided (eg. Age)
- **Single-valued attribute:** Attribute that has only a single value (eg. StudNr)
- **Multi-valued attributes:** Attributes that have many values (eg. Qualification)

Attributes (cont)

- **Multi-valued attributes** require creating either:
 - Several new attributes, one for each component of the original multivalued attribute (not ideal) OR
 - A new entity, composed of the original multivalued attribute's components which is then related to the original entity in a 1:M relationship
- **Derived attribute:** Attribute whose value is calculated from other attributes (eg. Age, if you have DOB)
 - Need not be physically stored in the DB
 - Can be derived using an algorithm/formula

Table 4.2 - Advantages and Disadvantages of Storing Derived Attributes

TABLE 4.2

ADVANTAGES AND DISADVANTAGES OF STORING DERIVED ATTRIBUTES

	DERIVED ATTRIBUTE	
	STORED	NOT STORED
Advantage	Saves CPU processing cycles Saves data access time Data value is readily available Can be used to keep track of historical data	Saves storage space Computation always yields current value
Disadvantage	Requires constant maintenance to ensure derived value is current, especially if any values used in the calculation change	Uses CPU processing cycles Increases data access time Adds coding complexity to queries

Relationships

- Associations between entities that always operate in both directions
- **Participants:** Entities that participate in a relationship
- Each relationship is identified by a name that describes it, using an active or passive verb.
- **Connectivity:** Describes the relationship classification
- **Cardinality:** Expresses the min and max number of entity occurrences associated with 1 occurrence of the related entity. Use the format (min,max). These limits cannot be directly implemented through the DB design, so are not always shown in ERD, but implied
- Connectivities and Cardinalities are established by business rules

Figure 4.7 - Connectivity and Cardinality in an ERD

FIGURE 4.7 CONNECTIVITY AND CARDINALITY IN AN ERD

